THE UNIVERSITY OF HONG KONG

BACHELOR OF ENGINEERING

DEPARTMENT OF CIVIL ENGINEERING

CIVL1113: ENGINEERING MECHANICS AND MATERIALS

8 December 2015 TIME: 14:30 – 17:30 (3 hours)

Answer **5 (FIVE)** questions.
All questions carry equal marks.
Marks for sub-questions (if any) are given.

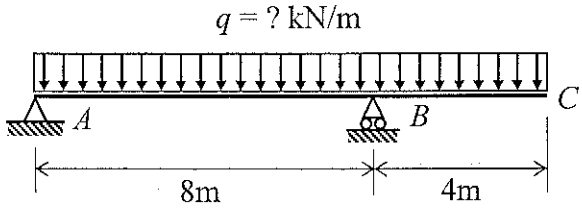
Do not use pencil!

Use of Calculators:

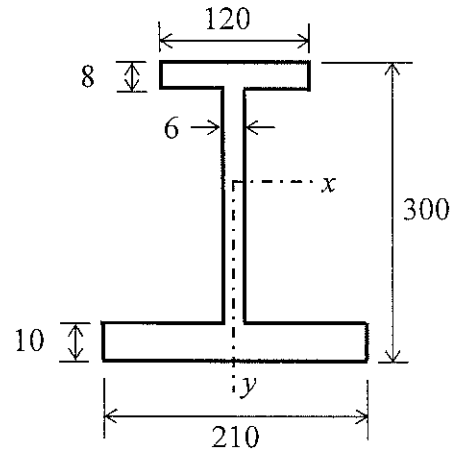
Only approved calculators as announced by the Examinations Secretary can be used in this examination. It is the candidates' responsibility to ensure that their calculator operates satisfactorily, and candidates must record the name and type of the calculator used on the front page of the examination script.

Section A

1. A simply supported beam with overhang (beam *ABC* in Figure 1) has span and overhang lengths of 8m and 4m, respectively. The beam is subjected to a uniformly distributed load (UDL) of q kN/m. The beam has an I-shaped cross section with unequal top and bottom flanges as shown in Figure 1. The beam has also unequal allowable strengths in compression and tension, i.e.: compression: $\sigma_c = 150$MPa; and
tension: $\sigma_t = 200$MPa.
Calculate the maximum value of UDL q in kN/m that can be carried by the beam.
(20 marks)



Beam



Cross section
Not to scale.
All dimensions in mm.

Figure 1

2. Sketch on the empty space provided the bending moment diagram (BMD), shear force diagram (SFD) and axial force diagram (AFD), if asked, for the structures listed in Figures 2a, 2b and 2c. The values are not required, but show clearly within the same diagram which values are identical, larger or smaller. Draw the BMD on tension side.

   (a) Sketch the AFD, BMD and SFD for the simply supported rigid frame *ABCD* shown in Figure 2a. The frame is subjected to a horizontal point load along *BC*. **(9 marks)**

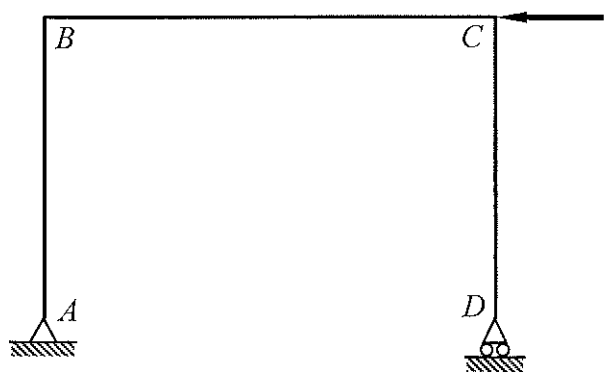
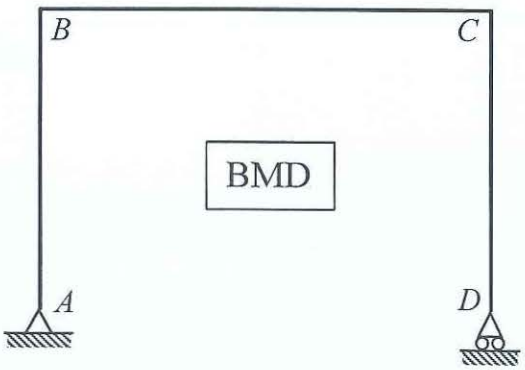
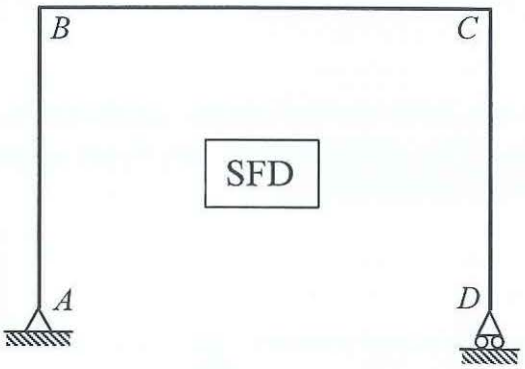


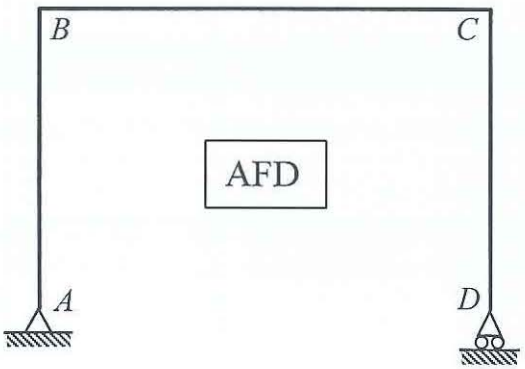
Figure 2a

University No.: .....

Do the answer for Question 2(a) below:







University No.: .....

2. (b) Sketch the BMD and SFD for the simply supported beam *ABC* shown in Figure 2b. A uniformly distributed load is applied along *AB* and *AB* > *BC*. **(5 marks)**

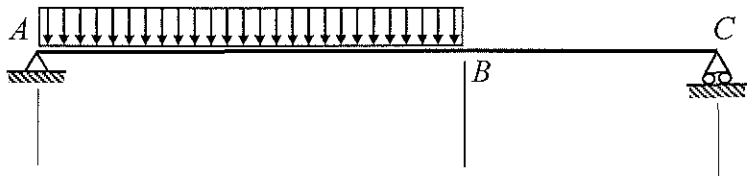
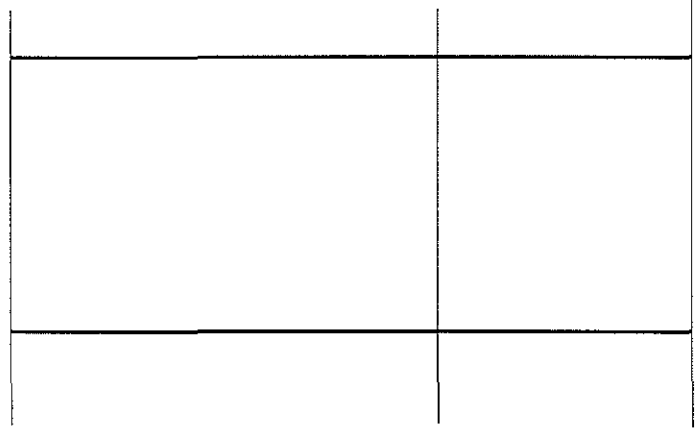


Figure 2b

Do the answer for Question 2(b) below:



BMD

SFD

2. (c) Sketch the BMD and SFD for the simply supported beam *ABCD* with overhangs shown in Figure 2c. Two point loads P_1 and P_2 are applied respectively at *A* and *D*. $P_1 < P_2$ and $a < b < L$. **(6 marks)**

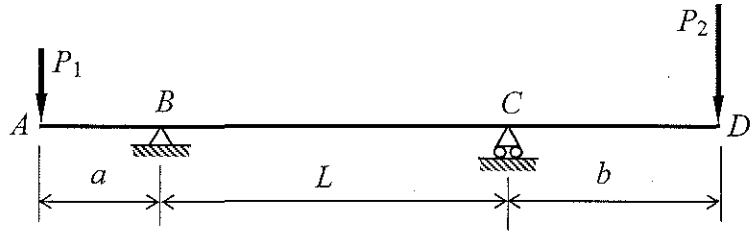
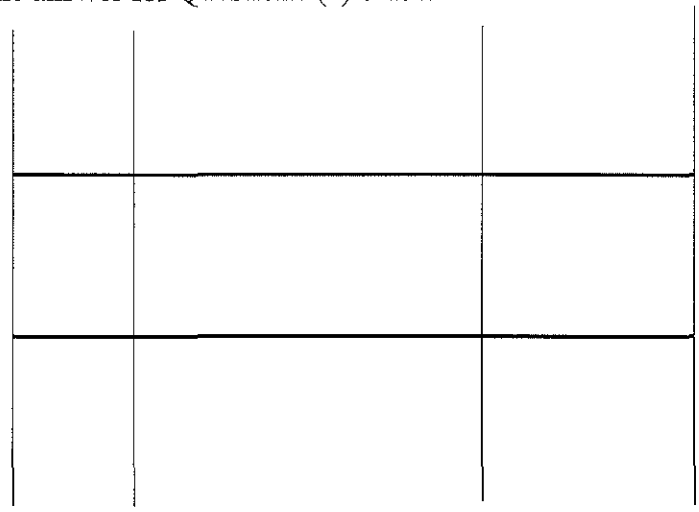


Figure 2c

Do the answer for Question 2(c) below:



BMD

SFD

3. Column AD having a symmetrical I-shaped cross section is subjected to an axial compressive load P as illustrated in Figure 3. The column is unrestrained about its major axis for the whole span length, except at its both supports. About its minor axis, the column is additionally retrained at every one third point of the span length (i.e. at B and C). If the Young's modulus of the column is E = 40GPa, calculate the maximum critical load P that can be resisted by the column. **(20 marks)**

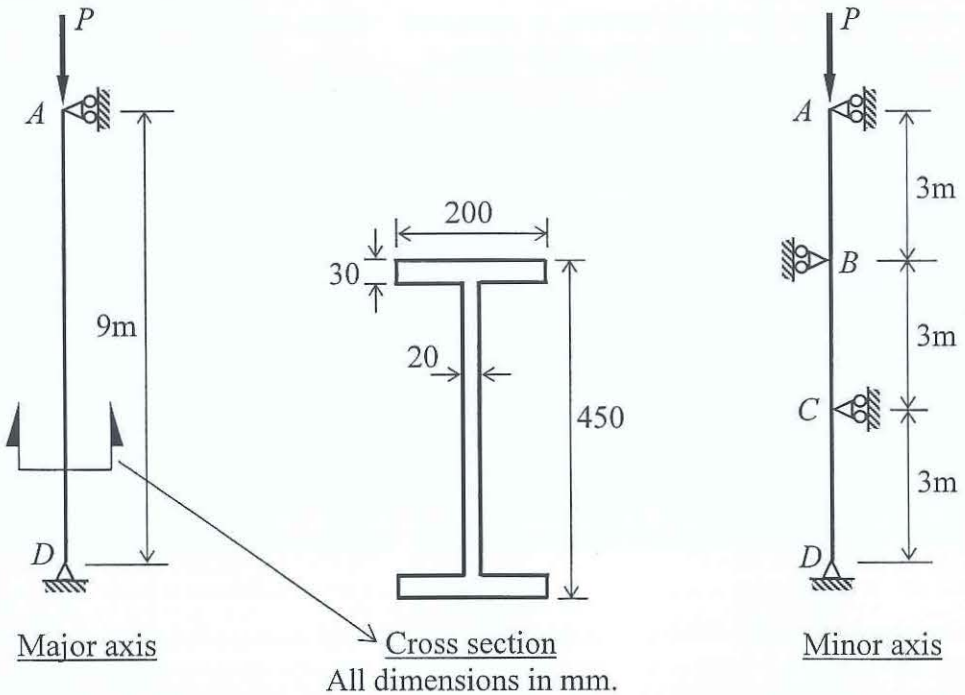


Figure 3